

Notes: Hsu (JF, 2004)

What Do Entrepreneurs Pay for Venture Capital Affiliation?

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1 Big picture

- **Question.** Do entrepreneurs value affiliation with high-reputation venture capitalists, and if they do, what price do they pay for that affiliation?
- **Core contribution.** The paper studies a market for affiliation. Entrepreneurs choose among competing financing offers, and venture capitalists differ in reputation and value-added services.
- **Empirical innovation.** The paper uses a rare survey sample of start-ups that received multiple Series A financing offers. This allows the author to hold the start-up fixed and compare competing investors' offers for the same firm.
- **Key idea.** A financing offer is not just money. A VC brings certification, industry experience, recruiting networks, supplier and customer contacts, investment-banking connections, and other extra-financial services.
- **Main choice result.** Offers from more reputable VCs are more likely to be accepted, even when the financial valuation offered is lower.
- **Main pricing result.** High-reputation VCs appear to acquire start-up equity at a discount. In the main fixed-effect valuation regressions, high industry deal experience is associated with an 10–14 percent valuation discount.
- **Interpretation.** Entrepreneurs pay for VC affiliation by accepting lower valuations in exchange for certification and services.
- **Important tension.** Cross-sectional regressions can give the opposite impression, because high-reputation VCs are associated with high-quality ventures and high absolute valuations. The multiple-offer, within-start-up design is needed to isolate the price of affiliation.
- **Takeaway.** Venture capitalists are differentiated inputs. The reputation of the investor changes both the entrepreneur's partner choice and the price at which equity is sold.

2 Incremental contribution of the paper

2.1 Contribution relative to VC contracting research

- Earlier VC work emphasized contracts, monitoring, staged financing, control rights, and incentive alignment between investors and entrepreneurs.
- Hsu shifts attention to the identity of the investor. The central object is not only how VC contracts are designed, but which VC the entrepreneur chooses and how much equity price concession is offered to that VC.
- The paper shows that investor heterogeneity is economically important at the contracting stage: the same start-up may receive several offers, and the offer from the most reputable investor need not be the highest-valuation offer.

Interpretation: The incremental contribution is to make VC identity and reputation central to the economics of early-stage financing, rather than treating VC money as homogeneous capital.

2.2 Contribution relative to certification literature

- Prior certification research often documents the benefits of affiliating with high-status intermediaries, such as prestigious underwriters or reputable investors.
- Much less evidence exists on the *cost* of certification.

- Hsu directly estimates that cost by examining competing offers to the same start-up. The valuation discount paid to reputable VCs is an economic price of certification.

Interpretation: This paper adds the missing demand-side price: entrepreneurs pay for certification and value-added affiliation by accepting worse cash-equity terms.

2.3 Contribution relative to empirical strategy in entrepreneurial finance

- Cross-sectional comparisons of funded start-ups are confounded because high-reputation VCs tend to invest in better start-ups.
- The multiple-offer sample allows start-up fixed effects, meaning all unobserved start-up quality is held constant.
- This design isolates differences across VC offers for the same firm.

Interpretation: The methodological contribution is powerful: use competing offers to convert an otherwise cross-sectional selection problem into a within-start-up comparison.

3 Data and measurement

3.1 MIT survey and sample construction

- The sample comes from the MIT Survey of New High-Tech Ventures.
- The main sample contains 149 start-ups receiving Series A financing offers.
- Fifty-one start-ups received multiple financing offers, producing 148 observed offers in the multiple-offer sample.
- The firms are high-technology ventures, many of them founded during the late 1990s.
- The multiple-offer subsample is the key analysis sample because it allows start-up fixed effects.

Interpretation: The data are rare because they observe offers that were declined as well as accepted. Most venture capital datasets observe only completed financings.

3.2 Dependent variables

The paper studies two dependent variables:

1. Offer acceptance:

$$\text{OfferAccepted}_{ij} = 1 \quad \text{if start-up } i \text{ accepts investor } j\text{'s financing offer.} \quad (1)$$

2. Relative valuation offered:

$$\text{RelativeValuation}_{ij} = \frac{\text{PreMoneyValuation}_{ij}}{\max_k \text{PreMoneyValuation}_{ik}}, \quad (2)$$

where the denominator is the highest pre-money valuation offered to the same start-up.

Interpretation: Relative valuation is the key price variable because it compares each VC's offer with the best valuation available to the same entrepreneur.

3.3 VC reputation measures

The paper uses several proxies for VC reputation or extra-financial resources:

- **High industry deal experience:** equals one if the VC has above-median prior deal experience in the start-up’s industry segment.
- **Industry reputation rank:** the entrepreneur’s rating of the VC’s industry reputation among the offers received.
- **High normalized funds raised:** equals one if prior funds raised per year of operation are above the sample median.
- **High network resources rating:** equals one if the VC receives the top rating on at least one network resource dimension, such as recruiting, customer/supplier contacts, or investment-banking contacts.
- **Boards per general partner:** the number of board seats per general partner, intended to capture time constraints or available partner attention.

Interpretation: These measures combine market-based reputation, entrepreneur-reported services, and capacity/resource measures. The paper’s main reputation proxy is high industry deal experience.

3.4 Financing controls

- **Angel investor:** equals one if the offer comes from an angel investor or angel group.
- **Corporate VC:** equals one if the offer comes from a corporate venture capital firm.
- **Financing offered:** total capital offered in the financing round, in millions of dollars.
- **Equity taken threshold:** equals one if the investor receives at least 30 percent of the start-up equity in the Series A round.

Interpretation: These controls help separate reputation effects from financing terms and investor type.

4 Empirical specifications

4.1 Offer acceptance regressions

The fixed-effects logit specification is:

$$\Pr(\text{OfferAccepted}_{ij} = 1) = F(\alpha_i + \beta_1 \text{Rep}_{ij} + \beta_2 \text{Valuation}_{ij} + \beta_3 \text{Terms}_{ij} + u_{ij}), \quad (3)$$

where α_i is a start-up fixed effect.

- The start-up fixed effect absorbs all characteristics of the venture that are constant across offers.
- Identification comes from differences in the reputation and terms of offers made to the same start-up.
- The main coefficient of interest is β_1 on VC reputation.

Interpretation: This specification tests whether entrepreneurs choose reputable VCs after comparing competing offers for the same company.

4.2 Valuation regressions

The valuation specification is:

$$RelativeValuation_{ij} = \alpha_i + \gamma_1 Rep_{ij} + \gamma_2 OfferAccepted_{ij} + \gamma_3 Terms_{ij} + \varepsilon_{ij}. \quad (4)$$

- The dependent variable is the valuation offered by investor j relative to the best valuation offered to the same start-up.
- A negative γ_1 means that reputable VCs offer lower valuations.
- Because start-up fixed effects are included, the coefficient is interpreted as a within-start-up discount.

Interpretation: This is the central price test. If entrepreneurs pay for affiliation, reputable VCs should be able to offer lower valuations and still be selected.

4.3 Selection correction

The paper also estimates fixed-effects Heckman-style models using the full sample of single-offer and multiple-offer start-ups. The first-stage selection equation predicts whether a start-up receives multiple offers. The second stage estimates offer acceptance or valuation in the full sample with selection controls.

Interpretation: The selection correction checks whether results depend on restricting analysis to the better firms that receive multiple financing offers.

4.4 Cross-sectional comparison

The paper also estimates cross-sectional OLS regressions using accepted offers only:

$$\log(PreMoneyValuation_i) = \alpha + \delta VCReputation_i + \Gamma X_i + \eta_i. \quad (5)$$

Interpretation: These regressions show why the within-start-up design matters. In cross-section, reputable VCs appear associated with higher valuations because they invest in better firms. This can obscure the valuation discount in competing offers.

5 Identification and empirical credibility

5.1 Main identification idea

- Start-up quality is difficult to observe and is strongly correlated with VC reputation.
- The multiple-offer sample lets the author compare several investors bidding for the same start-up.
- Start-up fixed effects absorb product quality, founder quality, market opportunity, and all other venture characteristics common to the offers.

Interpretation: The analysis identifies the value of VC affiliation from variation across offers, not across start-ups.

5.2 Selection concerns

- Start-ups receiving multiple offers are likely higher-quality ventures.
- Table I compares single-offer and multiple-offer start-ups and finds few statistically significant differences in start-up characteristics.
- Heckman robustness checks use the entire sample to address potential selection into multiple offers.

Interpretation: Selection is a concern, but the paper argues that selection into multiple offers is unlikely to explain the within-start-up reputation discount.

5.3 Measurement concerns

- Some reputation variables come from entrepreneur perceptions and may be retrospective.
- The paper therefore uses both subjective and objective reputation proxies.
- High industry deal experience is particularly important because it is based on Venture Economics data rather than only survey ratings.

Interpretation: The strongest evidence comes from objective VC experience measures paired with start-up fixed effects.

Table II
Descriptive Statistics of the Multiple Offers Sample

This table provides descriptive statistics of variables used in Tables III through VIII. The measure *VC offer accepted* is a dummy equal to 1 if an offer was accepted by the entrepreneur. The measure *pre-money valuation* is the product of the number of outstanding shares before the Series A round and the share price before the financing round (in \$ million). The measure *relative valuation offered* is the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The measure *industry reputation rank* is the rank (a ranking of 7 being first best, with lower values indicating lower reputation) of VC reputation among offers received, as rated by the entrepreneur. The measure *high-normalized funds raised* is a dummy variable equal to 1 if the number of prior funds (excluding buyout funds) the VC has raised per years of operation places the VC above the sample median. The measure *high network resources rating* is a dummy variable equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *boards per general partner* is the number of boards of directors per general venture capital partner. The measure *corporate VC* is a dummy variable equal to 1 if the VC is a corporate VC. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *financing offered* is the amount of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer.

Variable	Mean	SD	Source
Dependent Variables			
VC offer accepted	0.345	0.477	MIT Survey
Pre-money valuation	20.589	32.935	MIT Survey
Relative valuation offered	0.826	0.215	MIT Survey
VC Reputation Measures			
High industry deal experience	0.439	0.498	Venture Economics
Industry reputation rank	5.905	1.163	MIT Survey
High normalized funds raised	0.568	0.497	Venture Economics
High network resources rating	0.432	0.497	MIT Survey
Boards per general partner	4.198	2.405	Venture Economics
VC- and Financing-Term Controls			
Corporate VC	0.048	0.214	Corporate web sites
Angel investor	0.068	0.253	MIT Survey
Financing offered	7.863	8.944	MIT Survey
Equity taken threshold	0.520	0.501	MIT Survey

has become a standard measure in the literature (Gompers and Lerner (2000)).

Table II: descriptive statistics

Table III
Pairwise Correlation Matrix

Pairwise correlation matrix of the variables used in the empirical analysis. The measure *VC offer accepted* is a dummy equal to 1 if the offer was accepted by the entrepreneur. The measure *pre-money valuation* is the product of the number of outstanding shares before the Series A round and the share price before the financing round (in \$ million). The measure *relative valuation offered* is the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The measure *industry reputation rank* is the rank (a ranking of 7 being first best, with lower values indicating lower reputation) of VC reputation among offers received, as rated by the entrepreneur. The measure *high-normalized funds raised* is a dummy variable equal to 1 if the number of prior funds (excluding buyout funds) the VC has raised per years of operation places the VC above the sample median. The measure *high network resources rating* is a dummy variable equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *boards per general partner* is the number of boards of directors per general venture capital partner. The measure *corporate VC* is a dummy variable equal to 1 if the VC is a corporate VC. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *financing offered* is the amount of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
1	1											
2	-0.05	1										
3	0.05	0.40*	1									
4	0.13	0.26*	0.03	1								
5	0.43*	-0.22*	-0.04	0.12	1							
6	0.15	-0.25*	-0.03	-0.21*	0.06	1						
7	0.17*	0.27*	-0.09	0.08	0.07	-0.06	1					
8	0.12	0.10	-0.06	0.46*	0.25*	0.01	0.05	1				
9	-0.32*	0.02	0.02	-0.07	-0.06	-0.06	-0.07	-0.23*	1			
10	-0.10	-0.17*	-0.04	-0.24*	-0.05	0.24*	-0.02	-0.21	-0.06	1		
11	-0.05	0.73*	0.21*	0.15	-0.17*	-0.26*	0.33*	-0.02	-0.10	-0.19*	1	
12	0.13	-0.22*	-0.08	0.11	0.24*	-0.07	0.03	0.17	-0.10	-0.23*	0.09	1

t at the 5% level.

Table III: pairwise correlations

6.3 Table IV: univariate tests

Read:

- Accepted offers have lower average pre-money valuations than declined offers, though the difference is not statistically significant.
- Accepted offers have higher industry reputation rank and higher network resource ratings.
- Conditional means show that higher reputation measures are associated with lower relative valuations.

Economic message: Before controlling for start-up fixed effects, the data already suggest that entrepreneurs accept higher-reputation offers even when those offers are not the most generous valuations.

6.4 Table V: offer acceptance logit regressions

Read:

- The dependent variable is whether the entrepreneur accepts a given VC offer.
- High industry deal experience is positive and significant.
- High network resources rating is also positive and significant in the full specification.
- Relative valuation offered is not significant in the main fixed-effect logit specifications.

Economic message: Entrepreneurs appear to choose VCs based on reputation and resources, not only on the valuation offered.

6.5 Table VI: valuation regressions

Read:

Table IV
Univariate Difference in Means Tests

Panel A describes difference in means tests for accepted versus declined financing offers among offers received by start-ups with more than one Series A financing offer. The measure *pre-money valuation* is the product of the number of outstanding shares before the Series A round and the share price before the financing round (in \$ million). The measure *normalized industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment per years of operation places the VC above the sample median. The measure *normalized funds raised* is a dummy variable equal to 1 if the number of prior funds (excluding buyout funds) the VC has raised per number of years of operation places the VC above the sample median. The measure *industry reputation rank* is the rank (a ranking of seven being first best, with lower values indicating lower reputation) of VC reputation in the start-up's industrial segment among received offers, as rated by the entrepreneur. The measure *high network resources rating* is a dummy equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. Panel B describes conditional means of *relative valuation offered*, the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up, and equality of means tests for the upper and lower halves (divided at the median) of *normalized industry deal experience*, *normalized funds raised*, *industry reputation rank*, and *high network resources rating*.

Panel A: Difference in Means Tests			
	Offer Accepted	Offer Not Accepted	t-Stat: Equal Means
Pre-money valuation	17.733	22.090	0.764
Normalized industry deal exp.	3.844	2.810	-1.282
Normalized funds raised	0.667	0.515	-1.771*
Industry reputation rank	6.588	5.546	-5.711***
High network resources rating	0.549	0.371	-2.093**

Panel B: Conditional Means of Relative Valuation Offered			
	Top Half	Bottom Half	t-Stat: Equal Means
Normalized Industry Deal Exp.			
	0.812	0.844	0.880
Normalized VC Funds Raised			
	0.820	0.834	0.397
Industry Reputation Rank			
	0.798	0.843	1.233
High Network Resources Rating			
	0.804	0.842	1.070

*, **, or *** indicate statistical significance at the 10%, 5%, or 1% level, respectively.

Table V
VC Offer Acceptance Logit Regressions

This table shows start-up fixed-effects logit regressions on a sample of 148 offers across 51 start-ups receiving multiple Series A financing offers. Regression coefficients are reported, with standard errors in parentheses below. The dependent variable, *VC offer accepted* equals 1 if a financing offer was taken and 0 otherwise. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The measure *relative valuation offered* is the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. The measure *high network resources rating* is a dummy equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *corporate VC* is a dummy equal to 1 if the VC is a corporate VC. The measure *L financing offered* is the natural logarithm of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer.

Independent Variables	Dependent Variable = VC Offer Accepted N = 148 Observations		
	(5-1)	(5-2)	(5-3)
High industry deal experience	1.080** (0.512)	1.168** (0.532)	1.068** (0.556)
High network resources rating			1.415*** (0.519)
Relative valuation offered		0.606 (0.887)	2.157 (1.425)
Angel investor			0.558 (0.799)
Corporate VC			-0.450 (0.972)
L financing offered			-0.329 (0.593)
Equity taken threshold			1.015 (0.851)
Start-up fixed-effects	Yes	Yes	Yes
Prob > χ^2	0.028	0.070	0.016
Log likelihood	-49.195	-48.962	-42.974

** or *** indicate statistical significance at the 5% or 1% level, respectively.

- The dependent variable is relative valuation offered.
- High industry deal experience has a negative coefficient, around -0.10 to -0.14 .
- This implies that high-reputation VCs offer lower valuations relative to competing offers made to the same start-up.
- Financing offered and equity taken threshold also matter, with larger financing increasing relative valuation and higher equity taken reducing it.

Economic message: This table identifies the price of affiliation. Entrepreneurs pay for reputable VC affiliation by accepting a valuation discount.

Table VI
Valuation Regressions

This table shows the result of start-up fixed-effects OLS regressions on a sample of 148 offers across 51 start-ups receiving multiple Series A financing offers. The unit of observation is an offer to a start-up. Robust standard errors (adjusted for clustering by start-up firm) are reported (in parentheses). The dependent variable is *relative valuation offered*, the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The measure *VC offer accepted* equals 1 if a financing offer was taken and 0 otherwise. The measure *high network resources rating* is a dummy equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: Recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *corporate VC* is a dummy equal to 1 if the VC is a corporate VC. The measure *L financing offered* is the natural logarithm of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer.

Independent Variables	Dependent Variable = Relative Valuation Offered N = 148 Observations		
	(6-1)	(6-2)	(6-3)
High industry deal experience	-0.135** (0.057)	-0.144** (0.060)	-0.098*** (0.036)
High network resources rating			-0.048 (0.045)
VC offer accepted		0.026 (0.050)	0.041 (0.032)
Angel investor			-0.021 (0.084)
Corporate VC			0.016 (0.110)
L financing offered			0.261*** (0.054)
Equity taken threshold			-0.291*** (0.067)
Start-up fixed-effects	Yes	Yes	Yes
Constant	0.885*** (0.025)	0.880*** (0.027)	0.632*** (0.090)
Adj. R ²	0.098	0.094	0.460

** or *** indicate statistical significance at the 5% or 1% level, respectively.

6.6 Tables VII and VIII: robustness

Read:

- Table VII tests robustness of offer acceptance regressions using alternative reputation measures and Heckman selection corrections.
- Table VIII repeats analogous robustness checks for valuation regressions.
- The main acceptance result remains: reputation predicts offer acceptance.
- The main valuation result remains: high industry experience predicts lower relative valuation.
- Selection-correction models using the full sample produce similar results.

Economic message: The core results are not an artifact of one reputation proxy, one estimation method, or restricting attention to multiple-offer start-ups.

6.7 Table IX: cross-sectional regressions

Read:

- Table IX uses only accepted Series A financing offers and estimates cross-sectional OLS regressions of log pre-money valuation.
- High industry deal experience is positive in simpler specifications, consistent with high-reputation VCs backing higher-valued ventures.
- Once financing terms and start-up characteristics enter, the reputation coefficient weakens.

Table VII
Offer Acceptance Robustness Regressions

This table reports robustness checks on VC offer acceptance regressions. Fixed-effects logit regressions of *VC offer accepted*, a dummy equal to 1 if an offer was accepted by the entrepreneur, on the sample of multiple offers are found in columns 1–3. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The measure *high normalized funds raised* is a dummy variable equal to 1 if the number of prior funds (excluding buyout funds) the VC has raised per years of operation places the VC above the sample median. The measure *industry reputation rank* is the rank (a ranking of 7 being first best, with lower values indicating lower reputation) of VC reputation among offers received, as rated by the entrepreneur. The measure *high network resources rating* is a dummy equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *boards per general partner* is the average number of boards of directors per general venture capital partner. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *relative valuation offered* is the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. The measure *corporate VC* is a dummy equal to 1 if the VC is a corporate VC. The measure *L financing offered* is the natural logarithm of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer. Columns 4 and 5 show start-up fixed-effects Heckman sample selection probits on the entire sample of 246 offers across 149 start-ups receiving Series A financing offers. A first-stage selection (probit) equation determines the likelihood that a start-up receives multiple financing offers (*multiple offer*). The estimated covariates are: *L initial employees*, the natural logarithm of the pre-Series A number of start-up employees; *zero patents*, a dummy equal to 1 if the start-up does not have assigned patents; and dummy variables equal to 1 for each of the following industrial segments: *Internet* (infrastructure, retail, and service), *health sciences* (biotechnology and medical devices), and *computers* (software and hardware). Dummies for years of Series A investment (1998, 1999, and 2000) are also included. The term rho is the correlation between error terms of the selection and primary equations; lambda is the inverse Mill's ratio.

Independent Variables	Dependent Variable = VC Offer Accepted			
	FE Logits Multiple-Offers Sample (N = 148)		FE Heckman Linear Prob. Models, Entire Sample (N = 246)	
	(7-1)	(7-2)	(7-4)	(7-5)
High industry deal experience		1.070** (0.555)	0.328*** (0.120)	0.287** (0.119)
High-normalized funds raised	0.725* (0.437)			
Industry reputation rank		1.097*** (0.312)		
High network resources rating	1.428*** (0.518)	1.072*** (0.547)		0.375*** (0.104)
Boards per general partner		0.000 (0.001)		
Relative valuation offered	1.049 (1.360)	4.012** (1.779)	1.673 (1.385)	0.499* (0.277)
Angel investor	0.536 (0.792)	1.314 (0.864)	0.642 (0.811)	0.200 (0.186)

Table VIII
Valuation Robustness Regressions

This table reports robustness checks on *relative valuation offered*, the offered pre-money valuation to a start-up relative to the highest offered valuation received by that start-up. Fixed-effects OLS analysis (with robust standard errors) on the sample of multiple offers is reported in columns 1–3. The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median. The term *high normalized funds raised* is a dummy variable equal to 1 if the number of prior funds (excluding buyout funds) the VC has raised per years of operation places the VC above the sample median. The measure *industry reputation rank* is the rank (a ranking of 7 being first best, with lower values indicating lower reputation) of VC reputation among offers received, as rated by the entrepreneur. The measure *VC offer accepted* is a dummy equal to 1 if an offer was accepted by the entrepreneur. The measure *high network resources rating* is a dummy equal to 1 if a VC firm received the maximum Likert-scale measure (of 5) in at least one of the following entrepreneur-rated VC network resources: recruiting resources, contacts with customers and suppliers, or contacts with investment bankers. The measure *boards per general partner* is the number of boards of directors per general venture capital partner. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *corporate VC* is a dummy equal to 1 if the VC is a corporate VC. The measure *L financing offered* is the natural logarithm of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer. Columns 4 and 5 show start-up fixed-effects Heckman sample selection probits on the entire sample of 246 offers across 149 start-ups receiving Series A financing offers. A first-stage selection (probit) equation determines the likelihood that a start-up receives multiple financing offers (*multiple offer*). The estimated covariates are: *L initial employees*, the natural logarithm of the pre-Series A number of start-up employees; *zero patents*, a dummy equal to 1 if the start-up does not have assigned patents; and dummy variables equal to 1 for each of the following industrial segments: *Internet* (infrastructure, retail, and service), *health sciences* (biotechnology and medical devices), and *computers* (software and hardware). Dummies for years of Series A investment (1998, 1999, 2000) are also included. The term rho is the correlation between error terms of the selection and primary equations; lambda is the inverse Mill's ratio.

Independent Variables	Dependent Variable = Relative Valuation Offered				
	FE OLS Regressions Multiple-Offers Sample (N = 148)			FE Heckman's Entire Sample (N = 246)	
	(8-1)	(8-2)	(8-3)	(8-4)	(8-5)
High industry deal experience			−0.073* (0.039)	−0.135*** (0.044)	−0.098*** (0.034)
High normalized funds raised	0.061 (0.042)				
Industry reputation rank		−0.041** (0.018)			
High network resources rating	−0.056 (0.053)	−0.038 (0.047)			−0.048 (0.031)
Boards per general partner			−0.000* (0.000)		
VC offer accepted	0.056 (0.035)	0.046* (0.034)	0.031 (0.033)		0.041* (0.024)
Angel investor	0.114 (0.162)	−0.044 (0.083)	−0.064 (0.072)		−0.021 (0.054)

Table VII—Continued

Independent Variables	Dependent Variable = VC Offer Accepted				
	FE Logits Multiple-Offers Sample (N = 148)			FE Heckman Linear Prob Models Entire Sample (N = 246)	
	(7-1)	(7-2)	(7-3)	(7-4)	(7-5)
Corporate VC	−0.614 (0.920)	−0.795 (1.137)	−0.395 (0.969)		−0.051 (0.186)
L financing offered	−0.009 (0.574)	−1.242* (0.757)	−0.370 (0.600)		−0.039 (0.113)
Equity taken threshold	0.778 (0.840)	1.453 (1.127)	1.230 (0.800)		0.220 (0.154)
Log likelihood	−43.522	−35.754	−47.197	−249.406	−240.326
Rho				−0.000 (1.630)	−0.109 (0.709)
Lambda				−0.000 (0.735)	−0.046 (0.304)

*, **, or *** indicate statistical significance at the 10%, 5%, or 1% level, respectively. Heckman first-stage (selection) equation (SE in parentheses): *Multiple offer* = −0.262 (0.356) + 0.147 (0.108) *L init. emp.* − 0.461 (0.208) *Zero patents* + 0.663 (0.251) *Internet industry* + 0.278 (0.362) *Health science industry* + 0.315 (0.288) *Computer industry* − 0.199 (0.303) *Year 1998* − 0.080 (0.266) *Year 1999* + 0.292 (0.262) *Year 2000*

Table VIII—Continued

Independent Variables	Dependent Variable = Relative Valuation Offered				
	FE OLS Regressions Multiple-Offers Sample (N = 148)			FE Heckman's Entire Sample (N = 246)	
	(8-1)	(8-2)	(8-3)	(8-4)	(8-5)
Corporate VC	0.024 (0.093)	0.029 (0.114)	−0.014 (0.112)		0.016 (0.055)
L financing offered	0.355*** (0.082)	0.276*** (0.050)	0.287*** (0.046)		0.261*** (0.028)
Equity taken threshold	−0.290*** (0.074)	−0.283*** (0.071)	−0.306*** (0.065)		−0.291*** (0.039)
Constant	0.370*** (0.132)	0.793*** (0.148)	0.555*** (0.071)		
Adj. R ² /LL	0.593	0.471	0.497	−100.045	−57.314
Rho				0.000 (0.677)	−0.000 (0.960)
Lambda				0.000 (0.111)	−0.000 (0.118)

*, **, or *** indicate statistical significance at the 10%, 5%, or 1% level, respectively. Heckman first-stage (selection) equation (SE in parentheses): *Multiple offer* = −0.262 (0.356) + 0.147 (0.108) *L init. emp.* − 0.461 (0.208) *Zero patents* + 0.663 (0.251) *Internet industry* + 0.278 (0.362) *Health science industry* + 0.315 (0.288) *Computer industry* − 0.199 (0.303) *Year 1998* − 0.080 (0.266) *Year 1999* + 0.292 (0.262) *Year 2000*

Economic message: Cross-sectional regressions can mislead because high-reputation VCs sort into better ventures. The within-start-up offer comparison is necessary to estimate the cost of affiliation.

Table IX
Cross-sectional Regressions

This table reports OLS regressions, and is based on a sample of *accepted* Series A financing offers of 149 start-ups. The unit of observation is an offer to a start-up. The dependent variable is *L pre-money valuation*, the natural logarithm of the product of the number of outstanding shares before the Series A round and the share price before the financing round (in \$ million). The measure *high industry deal experience* is a dummy variable equal to 1 if the number of deals the VC has previously funded in the target start-up's industrial segment places the VC above the sample median in this category. The measure *boards per general partner* is the number of boards of directors per general venture capital partner. The measure *angel investor* is a dummy variable equal to 1 if a financing offer is from an angel investor or angel group. The measure *corporate VC* is a dummy equal to 1 if the VC is a corporate VC. The measure *L financing offered* is the natural logarithm of capital offered (in \$ million) by the investor. The measure *equity taken threshold* is a dummy variable equal to 1 if the investor receives at least 30% of start-up equity (the median of the sample) as a result of the Series A financing offer. The measure *L initial employees* is the natural log of the number of employees prior to receiving the Series A round. The measure *zero patents* is a dummy equal to 1 if the start-up does not have assigned patents. The measure *industry effects* represents a collection of indicator variables for the *Internet* (infrastructure, retail, and service), *health sciences* (biotechnology and medical devices), and *computers* (software and hardware) sectors (the communications industry is the excluded industry segment). Years of Series A investment dummies (1998, 1999, 2000) are also included.

Independent Variables	Dependent Variable = L Pre-money Valuation		
	(9-1)	(9-2)	(9-3)
VC characteristics			
High industry deal experience	0.478*** (0.145)	0.425*** (0.132)	0.058 (0.081)
Boards per general partner			-0.000 (0.000)
Angel investor			0.033 (0.113)
Corporate VC			-0.056 (0.173)
L financing offered			0.663*** (0.039)
Equity taken threshold			-0.753*** (0.072)
Start-up characteristics			
L initial employees		0.514*** (0.081)	0.189*** (0.050)
Zero patents		-0.223 (0.154)	-0.165** (0.083)
Industry effects		Yes	Yes
Year 1998		0.028 (0.215)	-0.157 (0.116)
Year 1999		0.263 (0.194)	-0.075 (0.105)
Year 2000		0.466** (0.197)	0.061 (0.107)
Constant	1.959*** (0.084)	0.828*** (0.253)	1.563*** (0.162)
R ²	0.070	0.338	0.820

** or *** indicate statistical significance at the 5% or 1% level, respectively.

7 Result synthesis

7.1 Entrepreneur choice

Entrepreneurs are more likely to accept offers from VCs with strong industry experience and network resources. This result supports the idea that the VC's nonfinancial services matter. It also means entrepreneurs trade off valuation terms against affiliation value.

7.2 Equity price of affiliation

The most important price result is that high-industry-experience VCs offer lower valuations. The discount is not a simple reflection of start-up quality because the regressions include start-up fixed effects. Thus, the discount is the price paid for affiliating with a reputable investor.

7.3 Certification versus value-added services

The paper cannot perfectly separate certification from operating value-added services. Industry experience and network resources can both certify quality and help the start-up succeed. The empirical result is therefore best interpreted as the total value of VC affiliation, combining certification and value-added resources.

7.4 Why multiple offers matter

Without multiple offers, one observes only the chosen VC and the final valuation. Since high-reputation VCs invest in better start-ups, a cross-sectional regression can show a positive relation between VC reputation and valuation. Multiple offers reveal that, conditional on the same start-up, high-reputation VCs can offer lower valuations.

8 Short summary

- The paper studies a rare sample of start-ups with multiple Series A financing offers.
- This data structure allows start-up fixed effects and within-start-up comparisons across VCs.
- Entrepreneurs are more likely to accept offers from more reputable VCs.
- High industry deal experience and network resource ratings are key predictors of offer acceptance.
- High-reputation VCs offer lower relative valuations.
- The valuation discount is roughly 10–14 percent in the main regressions.
- Robustness checks using alternative reputation measures and selection corrections support the main results.
- Cross-sectional regressions obscure the valuation discount because high-reputation VCs invest in high-quality start-ups.
- The paper shows that VC affiliation has a market price.

9 Conclusion

9.1 Core conclusion

Hsu shows that entrepreneurs pay for affiliation with reputable venture capitalists. They are more likely to accept financing offers from high-reputation VCs, even when these VCs offer lower valuations than competing investors. The discount reflects the market value of certification and nonfinancial services.

9.2 Economic intuition

Early-stage ventures lack hard assets and track records. Reputable VCs help solve this problem by certifying the start-up's quality and providing operating resources. Because these services are valuable and scarce, entrepreneurs are willing to sell equity at a lower price to obtain them.

9.3 Takeaway

The paper reframes VC financing as a market for affiliation, not just a market for capital. Entrepreneurial finance contracts contain both financial terms and reputational terms. Ignoring investor identity can lead to incorrect conclusions about the pricing of early-stage equity.